

COURSE TITLE : POLYMER MACHINERY & CAD LAB
COURSE CODE : 5097
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52/5
CREDITS : 2

TIME SCHEDULE

Module	Topics	Period
1	Basic drawing of polymer processing machinery	12
2	Perform computer aided design of 2D and 3D drawings .Design and draft various polymer products	20
3	Develop and draft various moulds and accessories of polymer products	20
TOTAL		52

COURSE OUTCOME

SI No.	Student will be able to
1	Identify the different parts and detailed drawing of various polymer processing Machineries
2	Perform computer aided design of 2D and 3D drawings .Design and draft various polymer products
3	Develop and draft various moulds and accessories of polymer products.

LIST OF EXPERIMENTS

MODULE I IDENTIFY THE DIFFERENT PARTS AND DETAILED DRAWING OF VARIOUS POLYMER PROCESSING MACHINERIES

- 1.1 Common usage assembly and detailed drawing of two roll mixing mill, Hydraulic Press and Calendars etc:
- 1.2 Drawing of factory lay out

MODULE II PERFORM COMPUTER AIDED DESIGN AND DRAFTING OF 2D AND 3D DRAWINGS.DESIGN AND DRAFTING OF VARIOUS POLYMER PRODUCTS

Introduction to computer aided drafting- main menu, starting a new drawing, drawing editor, Setting limits of drawing, grid, snap ,co-ordinate, ortho mode, zooming entering commands using mouse, pull down menu, getting help, data entry, and error correction. drawing line, arcs, circle ,erase ,undo ,oops , commands save and end command. Editing, Adding dimensions and text `Editing drawing using of various modified commands. Add dimensions and text on drawing hatching the sectional views. Developing simple orthographic views and dimensions it with test. Developing detailed orthographic view with all features -3D drawing-tools pallet-3D editing options-. Design and drafting of various polymer products like rectangular piece ,paper weight ,eraser ,doormat, Tea mat ,bushes ,gasket, paver mould. O-rings. blow moulding bottles.

MODULE III DESIGN AND DRAFTING OF RESPECTIVE MOULDS IN 3D.